Three stages of temporal interference

Draft22: prose and placement revision approved on 21 October 2026

We develop a mesoscale model of temporal-interference (TI) stimulation with three stages in series: membrane access, nonlinear detection of the difference frequency absent from the field, and circuit amplification. Access is represented by a short effective membrane time. We analyze detection and amplification in a Jansen–Rit column, an exact mean field of quadratic integrate-and-fire neurons, and its spiking network, and test spatial predictions in a realistic head. For weak fields, quadratic detection generates a beat proportional to the product of local carrier polarizations and a signed, DC-like steady source proportional to their summed squares. Transcranial alternating- and direct-current stimulation (tACS and tDCS) sources are linear in polarization to leading order. At the pinned Jansen–Rit operating point, 0.1–0.3 mV per TI carrier gives steady and beat sources approximately 2–5% of the tDCS and tACS sources when the direct input matches each carrier's polarization. Membranes slow relative to input add frequency dependence to detection. With a hippocampal montage, changing the current ratio moves envelope and mean-square foci oppositely but only rescales the beat map at fixed state. Near a Hopf bifurcation, resonant gain grows inversely with modal damping and is shared with tACS. At the same operating point, coupling changes gain independently of detection; amplification appears mainly in beat-locked timing, with little mean-rate change. At stronger fields the steady source alone can drive a gamma circuit across its oscillation threshold, as can an equal-power unmodulated carrier. Weak-field beat dose follows the polarization product rather than the envelope minimum. The steady source can act at depth, its sign set by detector curvature, not carrier polarity. Its magnitude has no strict interior maximum within regions of uniform conductivity, access, orientation, and detection. Polarization measurements, amplitude and frequency sweeps, and equal-power controls distinguish the stages and separate steady-source from beat-dependent effects.

The approved 300-word abstract is unchanged. This pass consolidates repetition and relocates procedures while retaining results and their local conditions. Giulio approved the joint plan and required the full PING system to stay in Methods. Draft20 (`e20c047`) remains the reading baseline; `24d34c1` is the immediate editing baseline. The canonical journal and supplement are rebuilt. The efficiency-bound block remains in its current location pending Giulio's placement decision. No simulation or numerical result changes.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Abstract and Introduction	A two-tone field has no difference-frequency line. Access creates local polarization; detection generates a beat and steady source; amplification determines the response. Retain the cellular and network literature, weak-perturbation scope, and approved FEM abstract sentence.	The reader learns the physical problem before the model equations. The tests distinguish the three stages rather than assigning every response to demodulation.
Theory: population curve and field coupling	Keep the JR sigmoid, synaptic response, and direct polarization. Derive the quadratic cross term and constant term together. Define and number the dynamic coefficient; retain its weak-input, memory, and nonzero-coefficient conditions.	The rate curve first explains what mixes the carriers. The NMM1 operator notation moves to the model appendix; it is not needed to understand this step.
Theory: envelope, spatial weighting, harmonics	Define the spatially varying envelope before using its harmonics. Retain the side-by-side map table, product/minimum relation, two-site example, fixed-pair current-ratio law, signed projections, and recurrent weighting. Keep the second-harmonic predictions, amplitude laws, and gain correction here; move the full harmonic ladder and mixed-detector inversion to supplementary Detection.	The reader can distinguish an envelope map from a detector output, then see how amplitude and harmonic measurements separate their predictions.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Theory: steady source	Retain the distinction between nonnegative mean-square polarization, signed source, and circuit response. Retain the convex/concave sign, polarity invariance, beat/DC inequality, 202-fold example, the quadratic subharmonic identity and all uniformity conditions. The envelope-norm proof and orientation averages move to supplementary Spatial maps. The direct tDCS/tACS comparison becomes a separate Theory subsection with the 2–5% result and its amplitude conventions.	The spatial restriction concerns the mean map under stated conditions; it does not prohibit an interior beat maximum. Orientation and equal-power qualifications stay here.
Theory: gain, access, operating point	Retain the synaptic attenuation estimates, resonant susceptibility, access–detection–gain product, static versus dynamic coefficients, access-time table, cable estimate, measured versus assumed times, and PEIX identities.	After detection is understood, the reader sees what suppresses the carriers and amplifies the beat, and why this does not establish physiological access.
Theory: exact mean field	Move the QIF introduction after the static argument. Keep the MPR equations in physical time and rate, its nonlinear equilibrium curve, threshold recruitment, and slow-synapse and carrier-relaxation conditions. Retain the signs and asymptotes of dynamic mixing.	QIF now answers a question already posed: when does a static curve suffice? A compact crossover derivation and exact coefficient belong in the article model appendix. The fixed-voltage coefficient and exponent catalogue remain in supplementary Detection. The static-rate relation and time-scale conditions stay in Theory; the repeated synaptic display moves to the article appendix.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Methods	State the purpose of waveform matching, lock-in, FEM geometry, and each model comparison before settings. Put projection choices with the head model. Place the shortened circuit schematic beside the first model explanation in Theory. Keep the full PING equations, effective input, observables, and convergence summaries. Move run catalogues and extended FEM procedures to topic-organized supplementary sections.	Methods connects each test to its observable. Detailed JR run settings, PING endpoint stability, and QIF cutoff results have explicit supplementary homes.
Results: JR and FEM	Combine the two-sentence beat demonstration with resonance. Retain the two-tone spectrum, curvature, square-law and linearization controls, kHz access checks, FEM maps and the sensitivity conclusion, pinned coupling sweep, timing ratios, and TI/tACS controls.	The evidence proceeds from beat generation to spatial consequences and recurrent amplification. A compact FEM table replaces the numerical catalogue. Sensitivity ranges and AM/two-tone comparison values move to supplementary tables; waveform identities and local limitations remain.
Results: NMM2 and spikes	Retain dynamic-source accuracy, the two-mode coupling curve, gain exponent and finite-amplitude deviations, attracting endpoint, spike cutoff qualification, both field amplitudes, equal-RMS carrier control, sidebands, and phase statistics.	The exact model confirms mixing and gain, then the network separates beat modulation from induced gamma. Numerical diagnostics are consolidated rather than repeated in captions.

Argument and retention map

Section order in the revised article and supplement

Place	Scientific content and destination	Contribution to the argument
Discussion	Group interpretation into access/mixing, spatial predictions, field-induced oscillations and gain, efficiency, population-model limits, and tests. Keep the Violante biological interpretation and conditions, removing repeated displacement values and relocating the ratio algebra. Move the Vieira numerical footnote to supplementary Efficiency while retaining its unmatched-input qualification. Keep the Cifra scope attached to its claim.	Each subsection answers one scientific question. The efficiency argument distinguishes generated drive from PLV and keeps the numerical ceiling within its stated family.
Predictions and Conclusion	Retain every test row, independent amplitudes, fixed-pair steering, state dependence, harmonic calibration, timing, blockade, and equal-power controls. Put the off-resonance measurement design beside these tests.	The conclusion states the supported findings and the measurements that separate mechanisms; detailed qualifications remain beside their claims.
Notation and supplement	Replace tilde variables with explicit physical units in the article and an explicit membrane-time convention in the uncoupled derivation. Keep all equations, citations, supplementary figures, LaNMM results, detector-bound derivation, and vocabulary metrics.	Article appendices contain notation, the JR specification, the slow-synapse display, and compact MPR crossover derivation. Supplementary sections contain access, detection, spatial maps, amplification, efficiency, laminar extension, and metrics/settings; figures accompany their topics. Article-appendix moves do not reduce the article count.

Changes and review conditions

No substantive scientific removal is proposed in this pass

The main article retains the PING system, core MPR system and static-rate curve, access evidence table, amplitude-matching conventions, and local cutoff and interpretation qualifications. The appendix supplies the JR specification and compact crossover calculation. Supplementary sections hold extended procedures, auxiliary derivations, sensitivities, and waveform checks. Each topic names the article claim it supports. Shared source blocks prevent divergence between editions.

The approved Violante shortening preserves its mean-map ordering, biological interpretation, and conditions. The Vieira numerical footnote moves with the comparison details. The efficiency-bound discussion and derivation remain in their existing narrative and supplementary locations pending Giulio's choice. Earlier CUT? items remain retained. The approved abstract, author order, figures, simulation caches, and numerical results are unchanged.

The first pass is recounted before further shortening is considered. Validation covers journal and supplement builds, retained labels and citations, notation, cross-document references, and rendered pages. Klaus checks the committed revision against the retention list. Public GitHub, Calliope, Zenodo, and further circulation remain held.